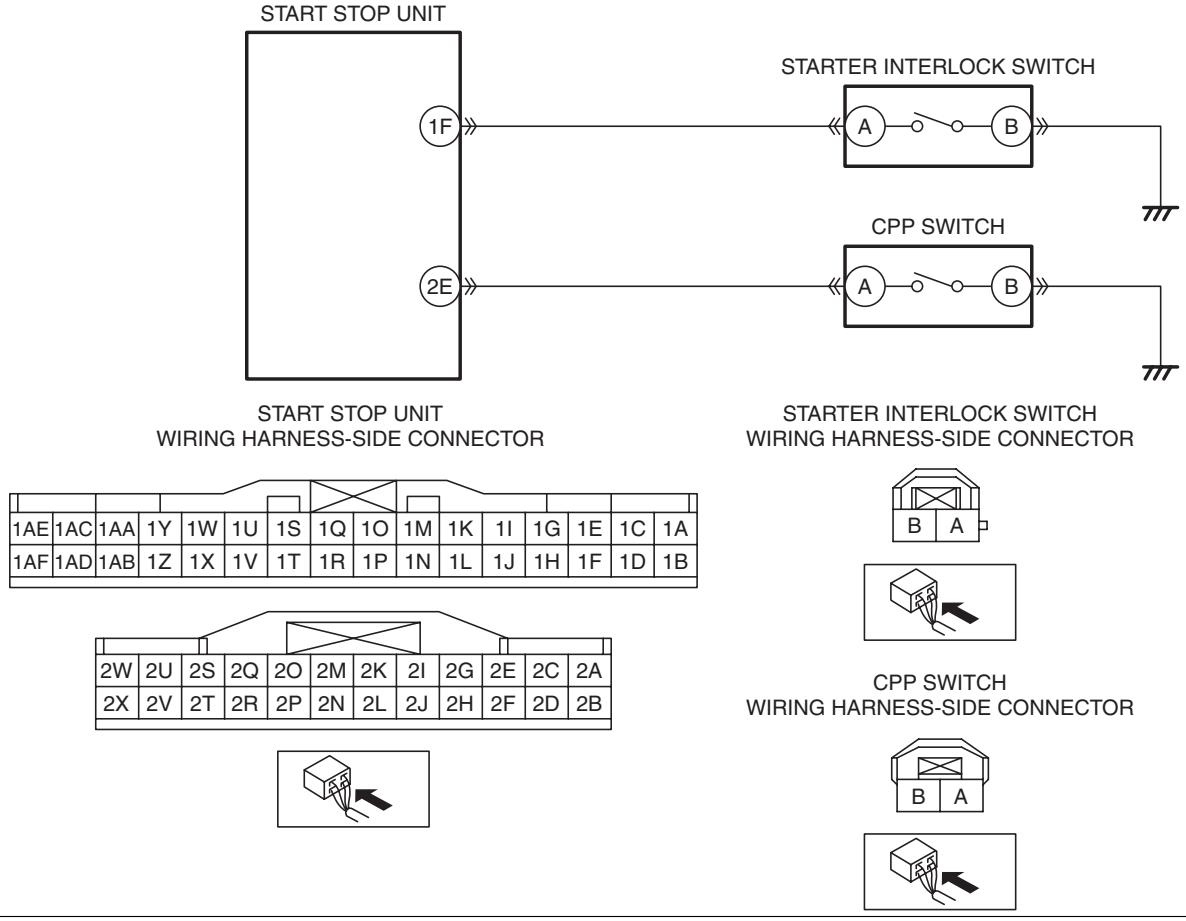
DTC P1708:29 [START STOP UNIT]

id0902p6023600

Description	Starter interlock switch circuit or CPP switch circuit malfunction
Detection condition	• Start stop unit detects that the starter interlock switch circuit is stuck on for **0.2 s or more**, or the CPP switch circuit is stuck off for **0.25 s or more**.
Fail-safe	• Change to the back-up mode. • Perform the control with starter interlock switch turned off. (Starter interlock switch circuit malfunction)
Possible cause	• Starter interlock switch connector or terminal malfunction • Starter interlock switch malfunction • Start stop unit connector or terminal malfunction • Short to ground in wiring harness between start stop unit terminal 1F and starter interlock switch terminal A • CPP switch connector or terminal malfunction • Open circuit in wiring harness between CPP switch terminal B and body ground • CPP switch malfunction • Open circuit in wiring harness between start stop unit terminal 2E and CPP switch terminal A • Start stop unit malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**PERFORM PID/DATA MONITOR INSPECTION AND DETERMINE MALFUNCTIONING LOCATION** • Display the following PIDs using the M-MDS. (See PID/DATA MONITOR INSPECTION [START STOP UNIT].) — CLUT_CUT_SW (Starter interlock switch) — CLUTCH_SW (CPP switch) • Are all PID values normal? (See PID/DATA MONITOR TABLE [START STOP UNIT].)	Yes	Go to Step 12.
		No	If the PID CLUT_CUT_SW value is not normal: • Go to the next step. If the PID CLUTCH_SW value is not normal: • Go to Step 6. If all PID values are not normal: • Go to Step 11.

Step	Inspection		Action
2	**INSPECT STARTER INTERLOCK SWITCH CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the starter interlock switch connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 12.
3	**INSPECT STARTER INTERLOCK SWITCH** • Inspect the starter interlock switch. (See STARTER INTERLOCK SWITCH INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See STARTER INTERLOCK SWITCH INSPECTION [SKYACTIV-D 1.5].) • Is the starter interlock switch normal?	Yes	Go to the next step.
		No	Replace the starter interlock switch, then go to Step 12. (See STARTER INTERLOCK SWITCH REMOVAL/INSTALLATION [F65M-R].) (See STARTER INTERLOCK SWITCH REMOVAL/INSTALLATION [F66M-R].)
4	**INSPECT START STOP UNIT CONNECTOR CONDITION** • Disconnect the start stop unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 12.
5	**INSPECT STARTER INTERLOCK SWITCH CIRCUIT FOR SHORT TO GROUND** • Verify that the start stop unit and starter interlock switch connectors are disconnected. • Inspect for continuity between starter interlock switch terminal A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between start stop unit terminal 1F and starter interlock switch terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 12.
		No	Go to Step 12.
6	**INSPECT CPP SWITCH CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the CPP switch connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 12.

Step	Inspection		Action
7	**INSPECT CPP SWITCH GROUND CIRCUIT FOR OPEN CIRCUIT** • Verify that the CPP switch connector is disconnected. • Inspect for continuity between CPP switch terminal B (wiring harness-side) and body ground. • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between CPP switch terminal B and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 12.
8	**INSPECT CPP SWITCH** • Inspect the CPP switch. (See CLUTCH PEDAL POSITION (CPP) SWITCH INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CLUTCH PEDAL POSITION (CPP) SWITCH INSPECTION [SKYACTIV-D 1.5].) • Is the CPP switch normal?	Yes	Go to the next step.
		No	Replace the CPP switch, then go to the Step 12. (See CLUTCH PEDAL POSITION SWITCH REMOVAL/INSTALLATION [F65M-R].) (See CLUTCH PEDAL POSITION SWITCH REMOVAL/INSTALLATION [F66M-R].)
9	**INSPECT START STOP UNIT CONNECTOR CONDITION** • Disconnect the start stop unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 12.
10	**INSPECT CPP SWITCH CIRCUIT FOR OPEN CIRCUIT** • Verify that the CPP switch and start stop unit connectors are disconnected. • Inspect for continuity between start stop unit terminal 2E (wiring harness-side) and CPP switch terminal A (wiring harness-side). • Is there continuity?	Yes	Go to Step 12.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between start stop unit terminal 2E and CPP switch terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 12.
11	**INSPECT START STOP UNIT CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the start stop unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to the next step.

Step	Inspection		Action
12	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].) • Switch the ignition ON (engine off or on). • Perform the work of depressing and releasing the clutch pedal **3 times**. • Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
13	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [START STOP UNIT].)
		No	DTC troubleshooting completed.